



# **Date and Time Management**

## **Practical Skills Module 3: Topic 4**

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## Practical Skills Module 3: Topic 4

|            |                                 |
|------------|---------------------------------|
| Topic Code | PM-03-PS04                      |
| Topic      | Declare Date, Time and Calendar |

### ***Scope of Practical Skill***

Given a range the learner must be able to:

- PA0401 Import date and time statements
- PA0402 Declare date objects in the main function
- PA0403 Get current date and time
- PA0404 Get current date
- PA0405 Print year, month, hour, minute and timestamp
- PA0406 Format data and time output with Strftime()
- PA0407 Create a datetime object from the given string with Strptime()
- PA0408 Suspend execution of the current thread for a given number of seconds with Sleep()
- PA0409 Format/edit the calendar
- PA0410 Set calculations for various task based on date, month, and year

### ***Applied Knowledge***

- AK0401 Concept, definition and functions of Python programming functionalities

### ***Internal Assessment Criteria***

- IAC0401 Expected results with Python date, time and calendar are achieved

## Resources:

| Knowledge                     | Information from Python Programmer Curriculum |
|-------------------------------|-----------------------------------------------|
| National Curriculum Framework | 900221-000-00-PM-03                           |



## Practical task

Follow the facilitator's instructions to complete the following activities:

Create a Python file named **event\_scheduler.py** and ensure it is stored in your task folder.

### Scenario: Event Management and Scheduling System

An event management company needs a Python program to handle event scheduling, manage date and time data, and generate user-friendly reports. The system should perform operations such as retrieving the current date and time, scheduling future events, formatting outputs for clarity, and delaying execution to enhance user experience or simulate event countdowns.

### Step 1: Import Required Modules

Import all the necessary Python modules to support date/time manipulation and calendar display. Use:

- `datetime` for working with dates and times.
- `time` for introducing execution delays.
- `calendar` for displaying monthly calendars.

### Step 2: Define the Main Function

Create a `main()` function as the central execution point of the program. All other operations should be called from here.

### Step 3: Display the Current Date and Time

Use `datetime.datetime.now()` to retrieve and print the full current date and time, including hours, minutes, and seconds.

### Step 4: Retrieve Only the Current Date

Extract and display only the current date (excluding the time) using `datetime.date.today()`.

### **Step 5: Extract Components from a DateTime Object**

Decompose the datetime object to print individual components such as:

- Year
- Month
- Day
- Hour
- Minute
- Second
- Timestamp (epoch time)

### **Step 6: Format Date and Time for Display**

Use the `.strftime()` method to format the date and time into a readable string format, such as:

- "Friday, 13 May 2025, 09:30 AM"

### **Step 7: Convert Strings into DateTime Objects**

Use the `.strptime()` method to parse a date/time string (e.g. "2025-06-30 15:00") into a valid datetime object.

### **Step 8: Simulate a Countdown or Loading Effect**

Use `time.sleep()` to pause program execution for a few seconds, simulating:

- A countdown to an event
- A loading or scheduling delay

### **Step 9: Display a Calendar**

Generate a calendar view for a given month and year using the `calendar.month()` function. This helps users visualise event schedules.

### **Step 10: Perform Date-Based Calculations**

Perform useful date operations such as:

- Calculating how many days remain until a scheduled event
- Computing the duration (difference in days) between two specific dates

## **Objective:**

By the end of this task, your Python script should serve as a functional **event scheduling and date management tool**, demonstrating real-world applications of date/time operations in Python.



### Take note

**Important:** Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.

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